

# Ritika Soni



## EDUCATION

| Examination | Degree / Institute                                                        | Year    |
|-------------|---------------------------------------------------------------------------|---------|
| Graduation  | B.Tech in Mathematics and Computing, Indian Institute of Technology Delhi | 2020-24 |

## WORK EXPERIENCE

- **Data Scientist** (*Fusion Data Intelligence Team*) | **Oracle** | **Bengaluru** Jul'24 - Present
  - **Scalable Metadata Ingestion and Search Platform**
    - \* Redesigned **source-format migration** by splitting business, schema & access-control metadata into separate flows, cutting future source-onboarding effort by about **70%** across similar integrations and repeat releases.
    - \* Replaced  **$O(m * n)$**  file-version indexing with  **$O(n)$  query-time merging** of version-specific indexes, reducing duplicate embeddings, storage, setup time & release-environment prep at scale.
    - \* Simplified rollout with **active-version checks**, safe cleanup rules & **zero-downtime index switching**, lowering deployment risk and preventing deletion of live search assets during redeployment and cleanup.
  - **NetSuite Intelligent Activation Engine** | *Patent in Progress*
    - \* Built **query-routing** and **candidate-generation engine** for NetSuite activation, splitting broad, pillar-level, transaction-based & free-text business requests into optimized retrieval paths for faster candidate selection.
    - \* Re-architected semantic-search-only retrieval into **hybrid lookup + embedding search**, reducing **LLM reranking** calls and cutting activation lookup latency by about **60-70%** for high-confidence transaction queries.
    - \* Built **transaction-aware activation logic** with contextual LLM reranking and one-to-many mapping, resolving backend analytics modules from business requests while reducing incorrect activation paths.
  - **NetSuite Source Onboarding for AI Concierge**
    - \* Led end-to-end onboarding of a new enterprise data source, NetSuite, into **AI Concierge** by mapping source-level metadata, module codes, access rules & transformation constraints before rollout.
    - \* Built adapter and **source-aware RAG/intent execution flows** to normalize metadata, generate missing access mappings at runtime, filter responses by permissions & create reusable integration pattern for future sources.
  - **Distributed Error Handling Framework**
    - \* Built error handling across prediction apps & AI Concierge, standardizing **exception propagation**, provider codes & customer-safe messages for PySpark, ADW, OCI GenAI, OSS, DAS & activation flows.
    - \* Designed granular classification for 20+ failure scenarios across service boundaries, separating service, application & recoverable failures; automated checks reduced manual triage by about **70%** across 5+ workflows.
- **Data Science Intern** (*Reinforcement Learning for Supplier Selection*) | **Oracle** | **Bengaluru** May'23 - Jul'23
  - Utilised RL methods - **Deep Q-Learning with Neural Networks** to determine optimal suppliers based on products
  - Generated **dynamic demand distributions** using RL, eliminating the need for pre-assumed distributions
- **Research Intern** (*Prof. Ganesh Neelakanta Iyer*) | **National University of Singapore** | **Singapore** Apr'22 - Jul'22
  - Used **unsupervised learning** for Exploratory Analysis of Geolocal Data using **Elbow Method** for best k value
  - Performed cluster formation using **K-Means**, formatted data with JSON, & visualized using **Folium & Geopy**
  - Forecasted number of Covid cases for next 10 days with supervised learning using SVM & Polynomial Regression

## PROJECTS

- **Voice Interview Coach** | **GitHub** | *Personal Project* Jun'26
  - Built a streaming voice agent (VAD, STT, LLM, TTS); clause-level synthesis cut response latency to 2.75s
  - Built full-duplex barge-in using semantic VAD, detecting turn-ends from intent rather than silence, backed by cancellable async tasks and floor-state tracking
  - Made the interviewer an agent: schema-constrained tool calls with LLM-judge evals against simulated candidates
- **Enterprise Work Intelligence Codex Plugin** | *OraHacks 2026* May'26
  - Built a **Codex plugin** that unifies approved context from Outlook, Slack, Jira, Confluence, Zoom summaries & code repositories into a prioritized **work-intelligence dashboard**
  - Designed **multi-agent execution workflow** that spawns independent sub-agents for parallel investigation, implementation planning, code analysis and source-backed Slack/email reply generation
  - Implemented compact **JSON context memory** persisting task history, source evidence, agent outputs and branch metadata, letting fresh Codex runs start with exact enterprise/repository context
- **Trajectory: Career Intelligence** | **GitHub** | *Crustdata-Y Combinator Hackathon* Apr'26
  - Built **Crustdata API-backed** accurate career analyzer with peer benchmarks, promotion velocity & career twins
  - Helps you find your **Career Twins** who had similar career trajectory as yours and then diverged to understand future career paths with pay bands, roles, warm intros & drafted outreach for networking
- **Counterspeech & Text Classification** | *Prof. Tanmoy Chakraborty* | *Natural Language Processing* Jan'24 - Apr'24
  - Fine-tuned **DialoGPT** with **PEFT adapters (LoRA & prompt tuning)** after T5/BART baselines for 5 intent classes
  - Built **BERT/GPT classifiers** on about 50K titles/abstracts using custom heads, preprocessing and 14-label training without external Word2Vec/GloVe embeddings

## TECHNICAL SKILLS

- **Languages:** Python, C/C++, SQL, Java
- **Tools & Frameworks:** PySpark, NumPy, Pandas, Scikit-Learn, PyTorch, Hugging Face Transformers, LangChain, LangGraph, RAG, Agentic Workflows, Git, Docker, Jenkins, OpenAI Codex, OCI, ADW, WebSockets, Speech-to-Text (Whisper / faster-whisper), Text-to-Speech, Voice Activity Detection (VAD), Semantic Turn Detection, LLM evaluation, MCP